MIL-C-23122B
7 April 1977
SUPERSEDING
MIL-C-23122A
26 November 1965

MILITARY SPECIFICATION

CONDENSERS, AIR COOLED, REFRIGERANT-12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers air cooled condensers for use with refrigerant-12.

1.2 Classification. The condensers shall be of the following types and classes (see 6.2):

Types:

Type I	-	22,000 Btuh (6.5 kW)
Type II	-	35,000 Btuh (10 kW)
Type III	-	46,000 Btuh (13 kW)
Type IV	-	57,000 Btuh (17 kW)
Type V	-	16,000 Btuh (5 kW)
Type VI	-	28,000 Btuh (8 kW)

Classes:

Class 1	-	Copper tubing, aluminum fins, aluminum housing
Class 2	-	Copper tubing, copper fins, aluminum housing
Class 3	-	Aluminum tubing, aluminum fins, aluminum housing
Class 4	-	Steel tubing, steel fins, steel housing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4130

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-P-664	-	Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-T-60	-	Tape: Packaging, Waterproof.

MILITARY

MIL-P-116	-	Preservation-Packaging, Methods Of.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-279	-	Compressor, Reciprocating, Power-Driven, Open-Type. For Use with Refrigerent-12.
MIL-STD-759	-	Condensers, Air Cooled, Refrigerant-12.
MIL-STD-773	-	Compressors, Hermetic: For Use with Refrigerant-12.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS LABORATORIES INC.

UL 207 - Refrigerant-Containing Components and Accessories, Nonelectrical

(Application for copies should be addressed to the Underwriters Laboratories Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, LI, NY 11746; 1655 Scott Boulevard, Santa Clara, CA 95050; or 2602 Tampa East Boulevard, Tampa, FL 33619.)

AMERICAN SOCIETY FOR HEATING, REFRIGERATING, AND AIR CONDITIONING ENGINEERS (ASHRAE)

Standard 20 - Methods of Testing for Rating Remote Mechanical-Draft, Air-Cooled and Evaporative Condensers

(Application for copies should be addressed to the American Society for Heating, Refrigerating and Air Conditioning Engineers, 345 East 47th Street, New York, NY 10017.)

* NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

* UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the contractor shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.3 and 6.3).

* 3.2 Codes and standards. The condensers shall comply with the applicable requirements of UL 207.

* 3.2.1 Compliance. Prior to approval of the first article, if one is submitted, or prior to commencing production, the contractor shall submit to the contracting officer or his authorized representative satisfactory evidence that the condensers he proposes to furnish under this specification, meet the applicable requirements of UL 207 as follows:

* 3.2.1.1 UL. Acceptable evidence of meeting the requirements of UL shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the condensers have been tested and conform to UL 207.

3.3 Materials. Materials used in the fabrication of the condenser shall be as shown in MIL-STD-759.

3.4 Design and construction. Condensers shall conform to the general and detail requirements of MIL-STD-759 and the additional requirements specified herein. The coil shall not leak when tested as specified in 4.5.1. The total heat rejection capacities of the condensers, expressed in Btuh, shall conform to MIL-STD-759 when tested as specified in 4.5.2.

3.4.1 Dehydration and sealing. The condenser tubes shall be internally dried by the manufacturer's standard process. The inlet and outlet connections of the tubes shall be sealed to prevent contamination and moisture accumulation during shipment and storage.

3.4.2 Fan. Unless otherwise specified (see 6.2), a fan shall be furnished with each condenser and the fan shall conform to MIL-STD-759.

3.4.2.1 Fan drive. When operating in conjunction with reciprocating compressors conforming to MIL-STD-279, the condenser fans shall be belt driven by the compressor or shall be direct electric motor driven. When operating in conjunction with hermetic compressors conforming to MIL-STD-773, the condensers shall have direct electric motor driven fans.

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3.5 Finish. Condensers shall be finished in accordance with the manufacturer's standard process.

3.6 Marking. Marking shall be in accordance with MIL-STD-130. The nameplate shall be fabricated of a corrosion resistant metal or a metal treated to resist corrosion. The nameplate shall be securely fastened in a conspicuous location. The nameplate data shall include manufacturer's identification and model number, type and class of condenser.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

* 4.3 First article. When a first article is required, the condenser shall be examined for defects listed in table I, dimensions specified in MIL-STD-759 and tested as specified in 4.5.2. The presence of any defect, any dimension not as specified, or failure of the test shall be cause for rejection of the first article.

* 4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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* 4.4.2 In-process inspection.

* 4.4.2.1 Process examination. Examination shall be made of the following fabrication processes to establish conformance with specified requirements: Dehydration and sealing (see 3.4.1). Whenever nonconformance is noted, corrections shall be made to the process and all items processed.

4.4.3 In-process testing.

4.4.3.1 Leakage. Each condenser coil shall be tested as specified in 4.5.1. Any nonconformance shall be cause for rejection of the applicable coil.

4.4.4 End item inspection. The inspection lot shall consist of all condensers of one type and class offered for inspection at one time. The sample unit shall be a completely fabricated condenser.

4.4.4.1 Visual examination. Examination shall be made of the condenser for the defects in table I. The inspection level shall be II with acceptable quality levels (AQLs) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not manufacturer's standard finish.		X
Construction and workmanship (general)	Not design specified, part missing, malformed, or broken.	X	
	Inlet and outlet connections not sealed.		X
Marking	Missing, incomplete, not as specified, or illegible.		X

4.4.4.2 Dimensional examination. Condensers shall be examined for defects in dimensions specified in MIL-STD-759. Any dimension not within specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

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4.4.4.3 End product testing. When a first article is not required, the initial production unit shall be tested as specified in 4.5.2 except in lieu of performing the test specified in 4.5.2 the contractor may furnish certified test data indicating that the capacity requirements of 3.4 have been met. Any nonconformance shall be cause for rejection of the unit.

* 4.4.5 Certificate examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.4.6 Packaging inspection. An examination shall be made to determine that packing and marking as required by section 5 are complied with. Defects shall be as specified in table II. The lot shall be the number of shipping containers of each item of the same type and class offered for delivery at the same time. The sample unit shall be one container fully prepared for delivery except that it need not be sealed. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

TABLE II. Inspection of packaging

Examine	Defect
Markings	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Component missing, damaged or otherwise defective.
Workmanship	Bulging or distortion of containers; components improperly secured.
Weight	Net weight exceeds requirements (applicable to fiberboard containers only).

4.4.7 Inspection of shipping containers. When shipping containers are required to comply with PPP-B-601 or PPP-B-636, examination for defects in the closure and reinforcing shall be in accordance with the appendix of the applicable container specification.

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4.5 Test methods.

4.5.1 Leakage. Each condenser coil shall be subjected to a gage pressure of 235 pounds per square inch (0.07 kg.M^2) using dry nitrogen gas, dry air, or refrigerant-12. The coil shall be submerged in water which is at standard room temperature and observed for leaks to determine compliance with 3.4. Any leaks shall constitute failure of this test.

4.5.2 Capacity. Condensers shall be tested for heat rejection capacity and airflow in accordance with ASHRAE Standard 20 to determine compliance with 3.4. The refrigerant gas temperature entering the condenser tubes shall be 135°F (57°C) saturated temperature and the exit refrigerant liquid temperature shall be a maximum 132°F (55.5°C). The minimum applicable air volume when using the applicable fan and fan motor specified in MIL-STD-759 and the air temperature through the condenser shall be as shown in table II of MIL-STD-759. Inability of the condenser to meet the capacity requirements of 3.4 shall constitute failure of this test.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Condensers equipped with direct motor driven fans shall be preserved and packaged in accordance with method III of MIL-P-116. Condensers equipped with belt driven fans shall be preserved and packaged in accordance with method I of MIL-P-116. Fan pulleys shall be coated with compound conforming to TT-P-664. All openings to the fan motor shall be sealed with tape conforming to type III or IV, class 1 of PPP-T-60.

5.1.2 Level C. Condensers shall be preserved and packaged to insure adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity. The contractor may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A packing. Condensers, preserved and packaged as specified in 5.1, shall be packed in shipping containers conforming to style RSC, grade V2s of PPP-B-636 or overseas type of PPP-B-601. Shipping containers conforming to PPP-B-601 shall be provided with skids as specified therein. As far as practicable, containers shall be uniform in shape and size and of minimum cube and tare consistent with the protection required. Each shipping container shall be closed and reinforced with flat steel strapping or tape banding in accordance with the appendix of the applicable container specification. The weight and dimensional limitations of the fiberboard shipping container shall not be exceeded.

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* 5.2.2.1 Weather-resistant fiberboard shipping containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

* 5.2.3 Level C packing. Condensers, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Physical protection. Internal anchoring, cushioning, blocking, and bracing shall be in accordance with MIL-STD-1186. The free-fall drop test shall be in accordance with appendix A of MIL-STD-1186.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

* 6.1 Intended use. The condensers are intended for use with Military refrigeration and air conditioning systems.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class of condenser required (see 1.2).
(c) When a first article is required (see 3.1).
(d) When fan is not required (see 3.4.2).
(e) Selection of applicable levels of preservation and packaging and packing (see 5.1 and 5.2).
(f) When weather-resistant class fiberborad shipping containers are required for level B shipment (see 5.2.2.1).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropiate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item, or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

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6.4 Changes from previous issue. The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-23122B CONDENSERS, AIR COOLED, REFRIGERANT-12

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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